

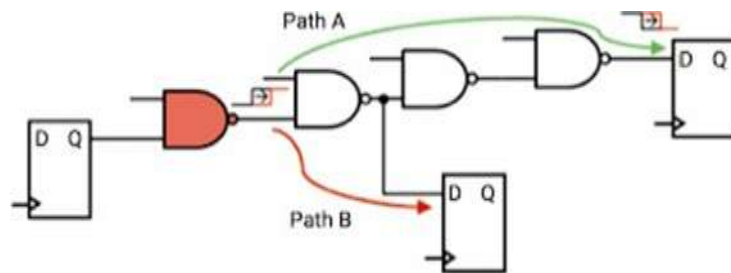
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|  <b>Marwadi University</b> | <b>Marwadi University</b><br><b>Faculty of Technology</b><br><b>Department of Information and Communication Technology</b> |                                  |
| <b>Subject: Digital VLSI Testing (01CT0629)</b>                                                             | <b>Aim: To detect delay faults in a full adder circuit.</b>                                                                |                                  |
| <b>Experiment No: 14</b>                                                                                    | <b>Date: 11-04-2026</b>                                                                                                    | <b>Enrolment No: 92301733056</b> |

**AIM:** To detect delay faults in a full adder circuit.

**IDE:** EDA Playground

### **THEORY:**

In digital circuits, path delay faults occur when signals take longer than expected to propagate through combinational paths, potentially leading to incorrect outputs. The critical path is the longest path that a signal can take from input to output, and it's crucial to ensure that this path meets the required timing constraints.



### **SYSTEMVERILOG CODE:**

#### **A. Full Adder Module**

```

module full_adder_delay(
    input A, B, Cin,
    output reg sum, Cout
);

    // Add small delays to simulate propagation delay
    always @(*) begin
        #2 sum = A ^ B ^ Cin;           // sum delay
        #3 Cout = (A & B) | (B & Cin) | (A & Cin); // carry delay
    end

endmodule

```

#### **B. Testbench**

```

`timescale 1ns/1ps

module tb_full_adder_delay;

```

|                                                                                                             |                                                                                                                            |                                  |
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```
reg A, B, Cin;
wire sum, Cout;
```

```
time start_time, end_time;
time delay_sum, delay_Cout;
```

```
// Instantiate DUT
full_adder_delay uut(
    .A(A),
    .B(B),
    .Cin(Cin),
    .sum(sum),
    .Cout(Cout)
);
```

```
// Task to check delay
task check_delay;
    input exp_sum, exp_Cout;
    input max_delay;
```

```
begin
    start_time = $time;
```

```
// Wait for output change
@(sum or Cout);
```

```
end_time = $time;
```

```
delay_sum = end_time - start_time;
delay_Cout = end_time - start_time;
```

```
// Logic check
if(sum !== exp_sum || Cout !== exp_Cout)
    $display("Logic Error: Expected sum=%b Cout=%b, Got sum=%b Cout=%b",
        exp_sum, exp_Cout, sum, Cout);
```

```
// Delay check
else if(delay_sum > max_delay || delay_Cout > max_delay)
    $display("Delay Fault: Delay=%0t exceeds %0t", delay_sum, max_delay);
```

```
else
    $display("Pass: sum=%b Cout=%b delay=%0t", sum, Cout, delay_sum);
end
endtask
```

```
// Test sequence
initial begin
    $dumpfile("dump.vcd");
    $dumpvars(0, tb_full_adder_delay);
```

|                                                                                                             |                                                                                                                            |                                  |
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```
$display("Full Adder Delay Fault Test Start");
```

```
// Initial values
```

```
A = 0; B = 0; Cin = 0;
```

```
#5;
```

```
// Test Case 1
```

```
A = 1; B = 1; Cin = 0;
```

```
check_delay(0, 1, 8);
```

```
#5;
```

```
// Test Case 2
```

```
A = 1; B = 0; Cin = 1;
```

```
check_delay(0, 1, 8);
```

```
#5;
```

```
// Test Case 3
```

```
A = 1; B = 1; Cin = 1;
```

```
check_delay(1, 1, 8);
```

```
#5;
```

```
// Test Case 4
```

```
A = 0; B = 1; Cin = 0;
```

```
check_delay(1, 0, 8);
```

```
#5;
```

```
$display("Testing completed");
```

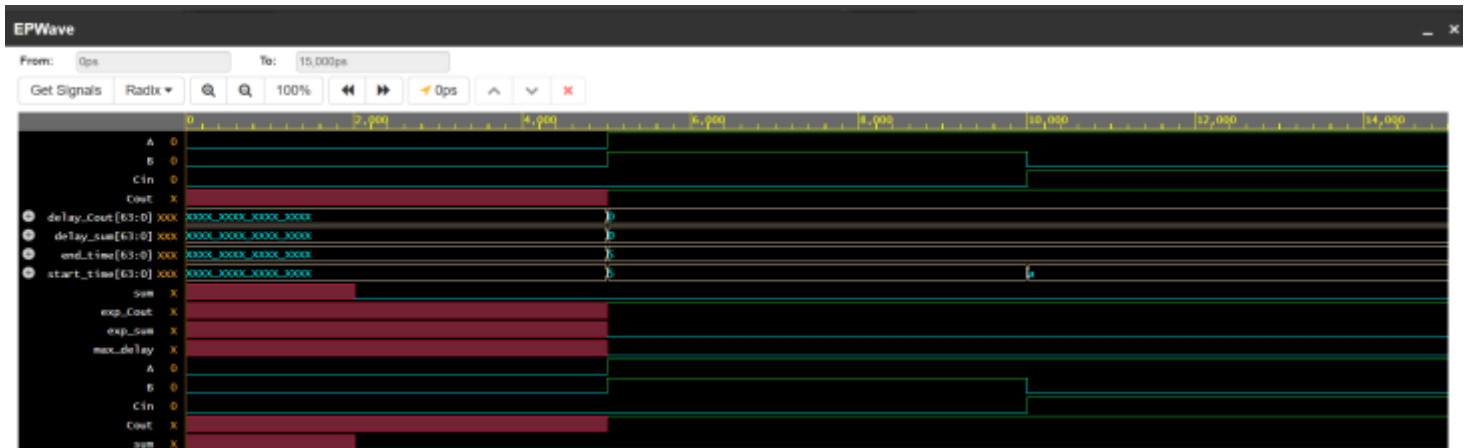
```
$finish;
```

```
end
```

```
endmodule
```

|                                                                                                             |                                                                                                                            |                                  |
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### SIMULATION RESULTS:



### CONCLUSION:

The experiment successfully demonstrated the detection of delay faults in a full adder circuit by applying appropriate input test patterns and observing the output response under timing constraints. It was observed that even when the logical functionality is correct, delays in signal propagation can cause incorrect outputs at higher operating speeds. By analyzing the sum and carry outputs for different input transitions, delay-related faults such as slow-to-rise and slow-to-fall were identified. This highlights the importance of at-speed testing and proper timing analysis in ensuring the reliable performance of digital circuits like the full adder.

|                                                                                                             |                                                                                                                            |                                  |
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### Post Lab Exercise:

1. What is a delay fault in a combinational circuit like a full adder?

A **delay fault** in a combinational circuit like a full adder occurs when a signal takes **longer than expected to propagate** through the circuit, even though the logical output is correct. This can cause incorrect results when the circuit operates at high speed.

2. How is propagation delay measured in the full adder testbench?

In the full adder testbench, propagation delay is measured by recording the time difference between the change in input signals and the corresponding change in outputs (Sum or Cout), often using simulation time variables.

3. Why are gate delays (# delays) introduced in the full adder design?

Gate delays (# **delays**) are introduced in the full adder design to **model real hardware behavior**, as actual logic gates do not switch instantaneously. This helps in analyzing timing performance and detecting delay faults during simulation.

4. What condition indicates a delay fault in the simulation results?

A delay fault is indicated when the output (Sum or Cout) does not change within the expected time after the input transition, or changes later than the specified delay limit.

5. Which outputs of the full adder (Sum or Cout) are checked for delay faults and why?

Both Sum and Cout outputs are checked for delay faults because they are the primary outputs of the full adder, and any delay in these signals directly affects the correctness and timing of the circuit's operation.